FUTURE LAB

REGULATIONS

For the educational and competitive intensive
on programming unmanned aerial systems
UAS “Bumblebee”

1. General Provisions

1.1. These Regulations (hereinafter referred to as the “Regulations”) define the procedure for organizing and conducting an educational intensive with a competitive component focused on programming unmanned aerial systems (hereinafter referred to as the “Intensive”).

1.2. The Intensive (Autonomous Drone Missions) is conducted in the format of a **training competition**, combining lectures, hands-on practical sessions, and a final demonstration of participants’ skills.

1.3. Participants of the Bumblebee UAS competition are university students aged 17–21. A team consists of 1–2 students. The presence of a team supervisor at the event is optional.

1.4. The official name of the event: **Future Lab Drone Intensive**.

1.5. Dates and venue: **January 20–22**,
United Arab Emirates, Abu Dhabi, KEZAD Warehouses (KLP-4), Unit B14-04.

1.6. The working language of the Intensive is **English**.
Certain elements (software environments, interfaces, educational materials) may also be provided in English or Arabic.

1.7. Participation in the Intensive implies full acceptance of these Regulations, as well as consent to photo and video recording and the processing of personal data in accordance with applicable legislation.

1.8. Registration link:
https://docs.google.com/forms/d/e/1FAIpQLSd0DiZa4ejNN0dp6ruJD-ioFU6hlpcvUyUsdgAERrdk6nem0w/viewform?usp=dialog

2. Objectives and Goals of the Intensive

2.1. Objective

The objective of the Intensive is to develop and practically apply participants’ skills in controlling and programming the **Bumblebee Unmanned Aerial System** for performing applied tasks involving payload operations.

2.2. Goals

- training in the fundamentals of controlling the Bumblebee UAS;
- mastering basic and applied drone programming;
- developing skills in LED indication control;
- training in programming and operating payload systems;
- developing manual and autonomous drone control skills;
- assessing the level of knowledge acquisition through practical tasks and the final demonstration.

3. Participants

3.1. Participation is open to individual participants or teams (the format is determined by the organizers).

3.2. All participants are required to:

- complete a safety briefing;
- comply with the instructions of experts and judges;
- follow all rules for operating UAS and related equipment.

4. Intensive Format

4.1. The Intensive lasts **three days** and includes:

- **4 lectures**;
- **4 practical sessions**, each conducted immediately after the corresponding lecture;
- a **final demonstration task at the exhibition stand**.

4.2. Each practical session is **evaluated** and contributes to the participant's final result.

4.3. Training and competition take place **simultaneously**:
participants acquire theoretical knowledge and immediately apply it in practice under time constraints.

5. Intensive Program

Day	Activity			
January 20	**Lecture 1**	**Practical Session**	**Lecture 2**	**Practical Session**
	Time			
	15:00–16:00	*16:00–16:30*	*17:00–18:00*	*18:00–19:30*
	Fundamentals of Unmanned Aerial Systems Control. Bumblebee UAS	- Bumblebee UAS flight in a simulator; - practicing takeoff, landing, and wire docking in a virtual environment.	Fundamentals of Programming. LED Indicator Control and Programming	-powering on the Bumblebee UAS; -accessing the programming environment; creating basic scripts; - controlling LED strip colors.
	Assessment: Ability to access the Bumblebee software environment and activate the LED strip in a specific color designated by the organizers.			
January 21	**Lecture 3**	**Practical Session**	**Lecture 4**	**Practical Session**

<table>
<tr><td rowspan="4"></td><td colspan="4">Time</td></tr>
<tr><td>14:00–15:00</td><td>15:00–15:30</td><td>16:00–17:00</td><td>17:00–18:30</td></tr>
<tr><td>Payload Programming for the Bumblebee Unmanned Aerial System</td><td>Development and debugging of a script that installs a clamp on a wire.</td><td>Training Flights and Practical Skills Development</td><td>Manual flights at the venue after:

-successfully passing the simulator exam;

-completing the on-site safety briefing.</td></tr>
<tr><td colspan="4">Assessment: Correctness of the algorithm and task execution logic.</td></tr>
<tr><td>January 22</td><td colspan="4">Final Demonstration at the Exhibition Stand
Time: 12:00–15:00

Participants must sequentially perform the following tasks:

1. In manual mode — take off and land the drone on the wire.
2. In autonomous mode:
o activate the LED strip in a specified color;
o install a clamp on the wire;
o change the LED color after the clamp installation is completed.
3. In manual mode — return the drone to the starting position.

The time allocated for the demonstration is limited and determined by the organizers.</td></tr>
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6. Evaluation and Results

6.1. The final stage of the Intensive is the demonstration of acquired skills at the exhibition stand.

6.2. The following are subject to evaluation:

- completion of practical tasks after each lecture;
- correctness of software solutions;
- compliance with the final task algorithm;
- accuracy and safety of flight operations.

6.3. The final result is determined based on:

- the cumulative score of practical sessions;
- the successful completion of the final demonstration task.

7. Awarding

The Organizer provides prizes to participants who place in prize-winning positions in the manner and quantities determined by the Organizer. The prizes are intended to be material items (for example: a laptop, a tablet, smart glasses); however, the Organizer reserves the right to substitute specific models and the number of prizes with other material items.

8. Safety Requirements

8.1. All flights must be conducted **exclusively within the designated area** and under expert supervision.

8.2. It is prohibited to:

- perform flights without expert authorization;
- violate safety briefing requirements;
- connect a battery with propellers installed outside the flight zone;
- interfere with the activities of other participants.

8.3. In the event of safety violations, a participant may be suspended from the Intensive.

9. Final Provisions

9.1. The organizers reserve the right to make changes to the program while maintaining the overall structure and objectives of the Intensive.

9.2. All disputes are resolved by the expert commission.